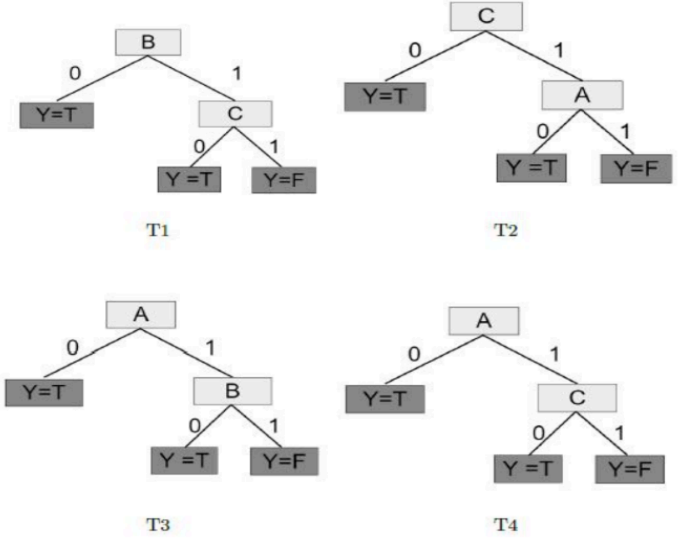
Week 7 | Solve With Instructors

Q1) Consider a training dataset with three features A, B and C and label Y as shown in the table below.

A	B	C	Y
0	0	0	T
1	0	0	T
1	1	0	T
1	1	1	F
0	1	1	T
0	0	1	T



Which of the following trees will give the zero mis-classification error on the training dataset? Note: In the trees, predictions are mentioned in the leaf nodes and in other nodes, the attribute along which the node is split is mentioned.

☐ $T1$ ☐ $T2$ ☐ $T3$ ☐ $T4$

Solution:

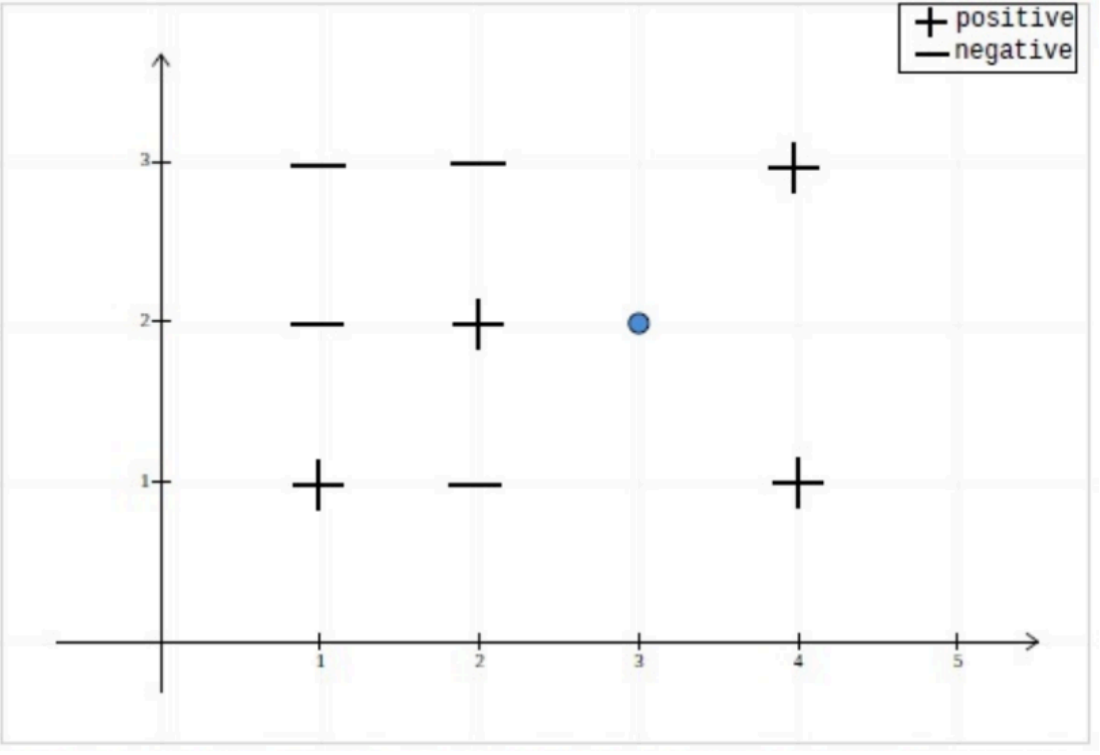
For $T1$, it miss-classifies the data point $\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T$. Hence it does not give zero mis-classification error.

For $T2$, it correctly classifies all data points. Hence it gives zero miss-classification error.

For $T3$, it miss-classifies the data point $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T$. Hence it does not give zero mis-classification error.

For $T4$, it correctly classifies all data points. Hence it gives zero miss-classification error.

Q2) Consider the following binary classification task for which we use $k - NN$ using the Manhattan distance.



All points are located at integer coordinates. If the test point (blue) is at $(3, 2)$, answer the given subquestions

a) How many points are at a distance of 2 from the test point?

Note: (1) We use the Manhattan distance. (2) We want those points that are exactly 2 units away from the test point.

Options :
(A) 2 points from the positive class
(B) 3 points from the positive class
(C) 2 points from the negative class
(D) 3 points from the negative class

b) What is the predicted label of the test point if $k = 1$? The label is 1 for the positive class and -1 for the negative class.

Solution: (a)

Data Point	Class of data point	Manhattan distance from $(3, 2)$
$(1, 1)$	$+$	3
$(1, 2)$	$-$	2
$(1, 3)$	$-$	3
$(2, 1)$	$-$	2
$(2, 2)$	$+$	1
$(2, 3)$	$-$	2
$(4, 1)$	$+$	2
$(4, 3)$	$+$	2

Hence 3 data points from negative class and 2 data points from positive class are at 2 Manhattan distance away from $(3, 2)$

(b)

Using above table, the sorted distance order of data point is :

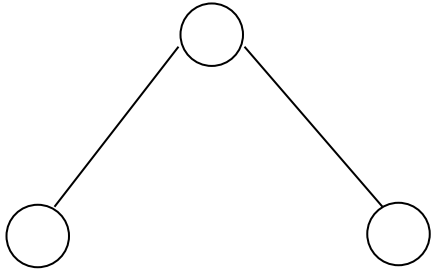
$$(2,2),(1,2),(2,1),(2,3),(4,1),(4,3)$$

So, by using only $k=1$, the closet neighbor is $(2,2)$, so the label of the test data point is $+$.

Q3) For a decision tree, each node has exactly two child nodes (balanced tree) . If the tree has a depth of 5, how many leaf nodes
are there?

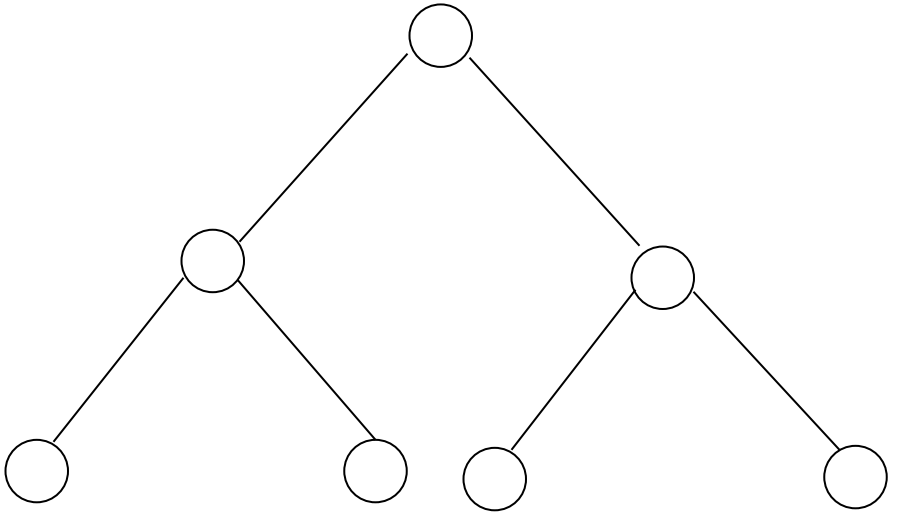
Solution:

When depth of tree is 1



we get 2 leaf nodes.

When depth of tree is 2



we get 4 leaf nodes.

So, the leaf nodes grows exponentially on power of 2. With depth 5. it would be $2^5 = 32$.

Q4) Which of the following options are **incorrect**?

(A) $k - NN$ algorithm can achieve zero mis-classification error with $k = 1$ on training dataset.
(B) Decision trees can achieve zero mis-classification error on training dataset.
(C) k is a hyperparmeter in $k - NN$ algorithm.
(D) Distance metric in $k - NN$ algorithm is a hyperparmeter.
(E) For $k = n$, the model will underfit in $k - NN$ algorithm where n is the total number of data points.
(F) Decision trees are sensitive to even a small change in a training dataset.
(G) Impurity of a pure leaf cannot be 1.
(H) In decision trees, to form a bounded region in a single dimension, we need a tree that has depth 2 at least.
(I) In $k - NN$ algorithm as dimensions increase, neighborhood information becomes less reliable.
(J) All of the above

Note: Consider that a data point is a neighbor of itself.

Solution:

(A) $k - NN$ algorithm can achieve zero mis-classification error with $k = 1$ on the training dataset.
- This is correct. With $k = 1$, each point is its own neighbor, so it will always classify correctly on the training data.

(B) Decision trees can achieve zero mis-classification error on the training dataset.
- This is also correct. Decision trees can perfectly fit the training data, potentially achieving zero error if they are allowed to grow without constraints.

(C) k is a hyperparameter in the $k - NN$ algorithm.
- Correct. k is indeed a hyperparameter that determines the number of nearest neighbors considered.

(D) Distance metric in $k - NN$ algorithm is a hyperparameter.
- Correct. The choice of distance metric (e.g., Euclidean, Manhattan) is also a hyperparameter in $k - NN$.

(E) For $k = n$, the model will underfit in $k - NN$ algorithm where n is the total number of data points.
- This is correct. If $k = n$, every point will be classified based on the majority label of the entire dataset, which leads to underfitting.

(F) Decision trees are sensitive to even a small change in the training dataset.
- Correct. Decision trees are sensitive to small changes, as they can lead to entirely different splits.

(G) Impurity of a pure leaf cannot be 1.
- Correct. In a pure leaf, all samples belong to the same class, so the impurity would be 0, not 1.

(H) In decision trees, to form a bounded region in a single dimension, we need a tree that has depth 2 at least.
- Correct. In a single dimension, (example a line) you need a depth of 2, for a bounded region.

(I) In $k-NN$ algorithm as dimensions increase, neighborhood information becomes less reliable.
- Correct. This is known as the "**curse of dimensionality**," where higher dimensions reduce the reliability of distance metrics in defining neighbors.

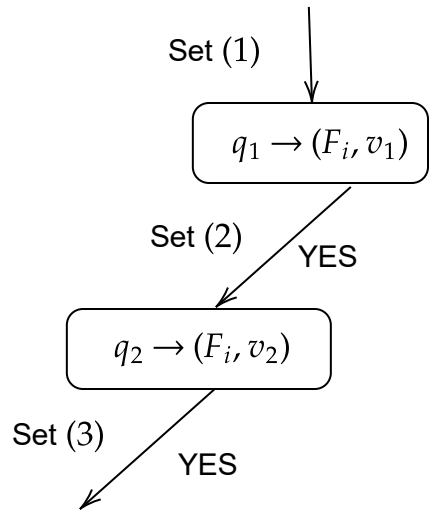
(J) All of the above
- Since all statements are correct, this option is incorrect.

Q5)

Set (k) is the set of all the data points which will encounter question k (q_k)

Select the possibilities for q_1 and q_2 so that $Set(3)$ is always an empty set

1. $q_1 \rightarrow F_i < v_1,\ q_2 \rightarrow F_i > v_2,\ v_1 > v_2$
2. $q_1 \rightarrow F_i > v_1,\ q_2 \rightarrow F_i < v_2,\ v_1 > v_2$
3. $q_1 \rightarrow F_i < v_1,\ q_2 \rightarrow F_i > v_2,\ v_1 < v_2$
4. $q_1 \rightarrow F_i > v_1,\ q_2 \rightarrow F_i < v_2,\ v_1 < v_2$



ANSWER : $2, 3$

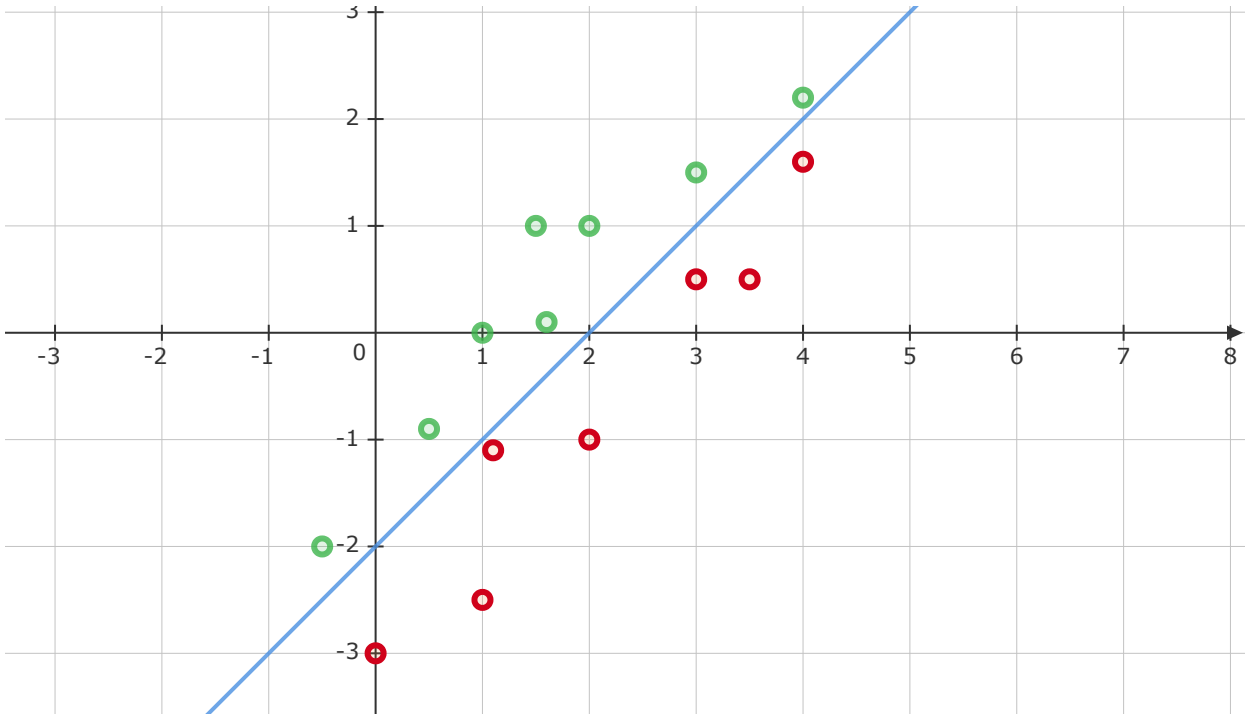
F_i is the feature in both the questions.

If $v_1 < v_2$, then the points satisfying $F_i < v_1$ will not satisfy $F_i > v_2$ and vice-versa.

Similarly, if $v_2 > v_1$, then the points satisfying $F_i > v_1$ will not satisfy $F_i < v_2$ and vice-versa.

For these two questions with the corresponding conditons, Set(3) will always be empty.

$Q6$) The datapoints given in the graph below are the training data points. Green refers to Class 0 and Red refers to Class 1.



Among the set of all linear functions, there exists a function $H_w(x_i) = sign\left(w^T x_i\right)$ such that the training error is 0. **TRUE or FALSE**

ANSWER : FALSE

If $H_w(x_i) = sign\left(w^T x_i\right)$, then $w^T x = 0$ is the it demarcator of the decision boundaries. $w^T x = 0$ will always pass through the origin. However, the blue line does not pass through the origin.

$Q7$) Set D has x data points. All the points in set D are made to answer question Q. The answer for Q can either be YES or NO. D_Y contains all the points in D for which the answer to question Q is YES and D_N contains all the points in D for which the answer to question Q is NO.

There are two labels : 0 & 1

There were 60 data points with labels as 1 in D and there are 36 data points with labels as 1 in D_Y.

There are 12 data points with labels as 0 in D_Y and there are 8 data points with labels as 0 in D_N.

Find the information gain due to question Q. Use log_2

ANSWER : 0

Data points with labels as 1 in

- $D \rightarrow 60$
- $D_Y \rightarrow 36$
- $D_N \rightarrow 24$

Data points with labels as 0 in

- $D \rightarrow 20$
- $D_Y \rightarrow 12$
- $D_N \rightarrow 8$

Total number of data points in

- $D \rightarrow 80$
- $D_Y \rightarrow 48$
- $D_N \rightarrow 32$

$$Entropy(p) = -(p\log_2(p) + (1-p)\log_2(1-p))$$

Here, $p \rightarrow$ Fraction of data points with labels as 1

Entropy for D
$p = 60/80 = 0.75$

$$-(0.75\log_2(0.75) + 0.25\log(0.25)) = 0.811$$

Entropy for D_Y
$p = 36/48 = 0.75$

$$-(0.75\log_2(0.75) + 0.25\log(0.25)) = 0.811$$

Entropy for D_N
$p = 36/48 = 0.75$

$$-(0.75\log_2(0.75) + 0.25\log(0.25)) = 0.811$$

Information Gain for question Q

$$Entropy(D) - (\gamma)Entropy(D_Y) - (1-\gamma)Entropy(D_N)$$

Here, $\gamma = \dfrac{|D_Y|}{D} = \dfrac{48}{80} = 0.6$

$$\begin{aligned} &Entropy(D) - (0.6)Entropy(D_Y) - (0.4)Entropy(D_N) \\ &= 0.811 - 0.6(0.811) - 0.4(0.811) \\ &= 0 \end{aligned}$$

Question 1

Consider a binary classification problem with a training dataset of 80 points, evenly distributed between two classes (40 points in each class). You decide to train a k-NN algorithm with $k = 3$. Each point is considered its own neighbor during classification.

a) What is the minimum number of misclassifications that can occur in the training dataset when using this k-NN algorithm?

b) Assuming there are outliers, the decision boundary becomes smoother with increasing value of k in a k-NN algorithm.

Solution to Question 1

a)Since $k = 3$ and each point is considered its own neighbor, each point has at least one neighbor from its own class. Therefore, the minimum number of misclassifications is achieved by considering only the nearest neighbors. If the dataset is well-separated, the minimum misclassification rate should be zero, as each point can be correctly classified by the nearest two neighbors.

b)True,Increasing the value of k in a k-NN algorithm does tend to make the decision boundary smoother, as it averages over more neighbors, which reduces the effect of outliers.

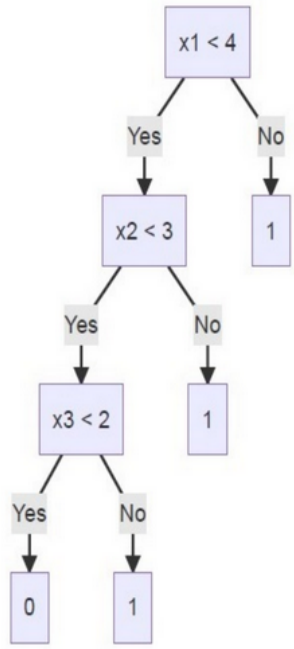


Figure 1: Decision tree

Consider a dataset in $\mathbb{R}^3$. Each data-point is represented by $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^T$. The features in this problem are all positive. That is, $x_1, x_2, x_3 > 0$ for all data-points. Consider the above shown decision tree trained on this dataset. The features are represented without the subscript in the nodes; for example, x_1 is represented as $x1$.

Consider only those points for which x_1, x_2, and x_3 are all positive. Let S be the set of all points in $\mathbb{R}^3$ that are predicted as 0 by this decision tree. What is the volume of the region S?

The region S is defined by the conditions:

$$x_1 < 4, \quad x_2 < 3, \quad x_3 < 2$$

Since each of these conditions represents an interval in $\mathbb{R}^3$, the volume of the region S can be calculated as the product of the lengths of these intervals:

$$\text{Volume of } S = 4 \times 3 \times 2 = 24$$

Thus, the volume of the region S is 24.

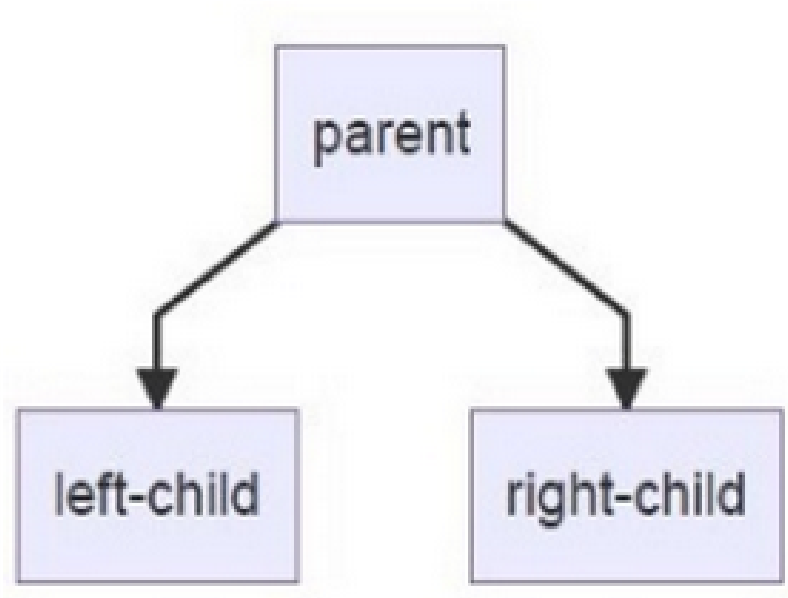


Figure 2:

Question:3

Consider a decision stump for a binary classification problem that has 500 data points at the parent node, out of which 200 data points go into the left child. The number of data points that belong to class 1 in the parent node is 300. The number of data points that belong to class 1 in the left child is 50. The labels are in $\{1, 0\}$.

Note for calculations: Use $\log_2$ for all calculations that involve logarithms. For all questions, enter your answer correct to three decimal places. Use three decimal places even while calculating intermediate quantities.

Based on the above data, answer the following questions:

1. What is the entropy of each node?
2. What is the information gain from splitting the parent node?

Answer

The entropy H of a node for a binary classification problem with probabilities p and $1-p$ for the two classes is given by:

$$H = -p\log_2(p) - (1-p)\log_2(1-p)$$

1. Entropy of the Parent Node

In the parent node, there are 500 data points, with 300 belonging to class 1 and 200 belonging to class 0. Thus, the probability of class 1, p_1, is:

$$p_1 = \frac{300}{500} = 0.6$$

The probability of class 0, p_0, is:

$$p_0 = \frac{200}{500} = 0.4$$

The entropy of the parent node is:

$$H_{\text{parent}} = -(0.6\log_2(0.6) + 0.4\log_2(0.4))$$

Calculating this, we get:

$$H_{\text{parent}} \approx -(0.6 \times -0.737) - (0.4 \times -1.322) = 0.974$$

2. Entropy of the Left Child

In the left child, there are 200 data points, with 50 belonging to class 1 and 150 belonging to class 0. Thus, the probability of class 1, p_1, is:

$$p_1 = \frac{50}{200} = 0.25$$

The probability of class 0, p_0, is:

$$p_0 = \frac{150}{200} = 0.75$$

The entropy of the left child is:

$$H_{\text{left}} = -(0.25\log_2(0.25) + 0.75\log_2(0.75))$$

Calculating this, we get:

$$H_{\text{left}} \approx -(0.25 \times -2) - (0.75 \times -0.415) = 0.811$$

3. Entropy of the Right Child

In the right child, there are 300 data points (500 - 200), with 250 belonging to class 1 (300 - 50) and 50 belonging to class 0. Thus, the probability of class 1, p_1, is:

$$p_1 = \frac{250}{300} \approx 0.833$$

The probability of class 0, p_0, is:

$$p_0 = \frac{50}{300} \approx 0.167$$

The entropy of the right child is:

$$H_{\text{right}} = -(0.833 \log_2(0.833) + 0.167 \log_2(0.167))$$

Calculating this, we get:

$$H_{\text{right}} \approx -(0.833 \times -0.263) - (0.167 \times -2.584) = 0.651$$

4. Information Gain

The information gain from splitting the parent node is given by:

$$\text{Information Gain} = H_{\text{parent}} - \left(\frac{200}{500} H_{\text{left}} + \frac{300}{500} H_{\text{right}}\right)$$

Substituting the values:

$$\text{Information Gain} = 0.974 - \left(\frac{200}{500} \times 0.811 + \frac{300}{500} \times 0.651\right)$$

Calculating each term:

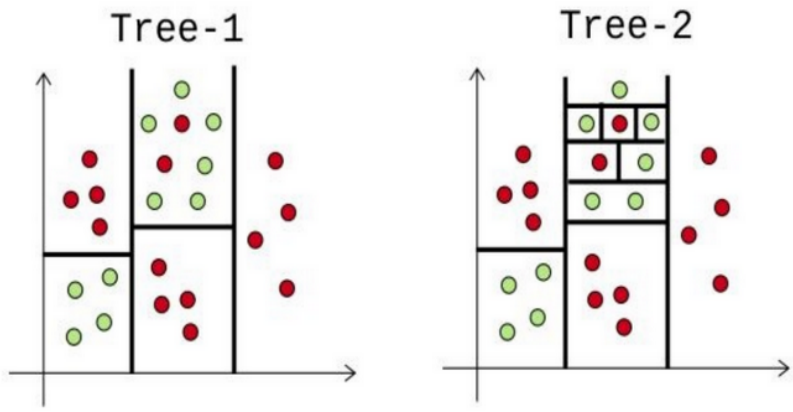
$$\frac{200}{500} \times 0.811 = 0.3244$$

$$\frac{300}{500} \times 0.651 = 0.3906$$

Thus,

$$\text{Information Gain} = 0.974 - (0.3244 + 0.3906) = 0.974 - 0.715 = 0.259$$

$$H_{\text{parent}} = 0.974, \quad H_{\text{left}} = 0.811, \quad H_{\text{right}} = 0.651, \quad \text{Information Gain} = 0.259$$



Question:4

Consider a training dataset for a binary classification problem . Two decision trees are trained on the same dataset.

The decision regions obtained are plotted for both the trees

a) Which of these two trees is likely to perform better on test data?

b)The training dataset for a binary classification problem has 100 points, 50 of which belong to class +1. Consider a k-NN algorithm with k = 1 that is used to predict the labels of the training data points. A point is considered as its own neighbor. Based on this setup, study the following statements

- **S1:** The number of points that are misclassified by the classifier is zero.
- **S2:** Since the training error is zero, we have found a very good classifier for this problem.

Answer a):

Tree-1 is likely to perform better on the test data. Here's the explanation:

Tree-1 has simpler and more general decision boundaries, while Tree-2 has very detailed and complex boundaries. Tree-2 seems to overfit the training data by creating very narrow and specific regions for each class. This means it likely captures noise in the training set, leading to poorer generalization on unseen test data.

On the other hand, Tree-1 has fewer splits and captures the general distribution of the classes without focusing too much on specific data points. As a result, Tree-1 is likely to generalize better on test data because it is less prone to overfitting.

In summary: - **Tree-1** is simpler and less likely to overfit, so it should perform better on the test data. - **Tree-2** has complex decision boundaries, indicating overfitting to the training data, which may lead to poor performance on test data.

Answer b):

Statement S1 is True.

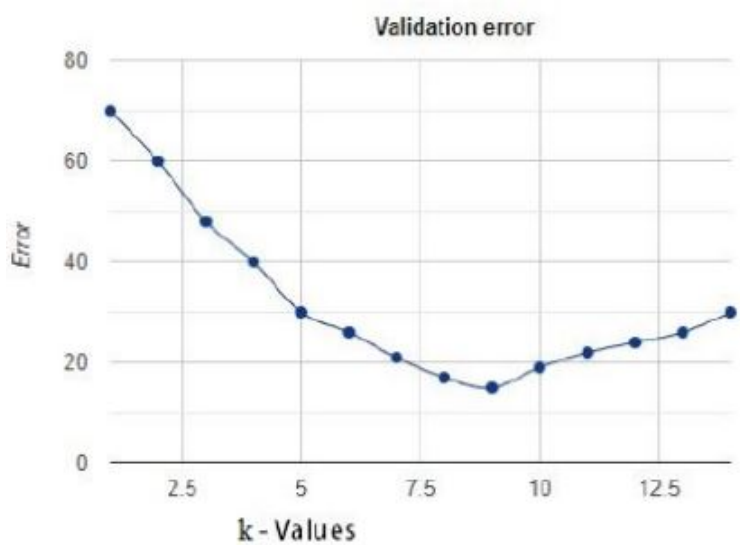
In a 1-NN (1-Nearest Neighbor) classifier, each point is classified based on the nearest neighbor, which, in this case, is the point itself. Since a point is its own neighbor, it will always match its own class label, resulting in zero misclassifications. Therefore, **S1 is correct**.

Statement S2 is False.

While the training error is zero due to the 1-NN classifier assigning each point its own label, this does not necessarily indicate a good classifier for the problem. A 1-NN classifier is very prone to overfitting because it memorizes the training data and may perform poorly on unseen test data. Zero training error does not guarantee good generalization to test data, which is essential for a good classifier. Therefore, **S2 is incorrect**.

Machine Learning Techniques
Week 7 (KNN, Decision Tree)
Sept 2023 term

1. What would be the best value for k to be used in KNN algorithm based on the graph given below?



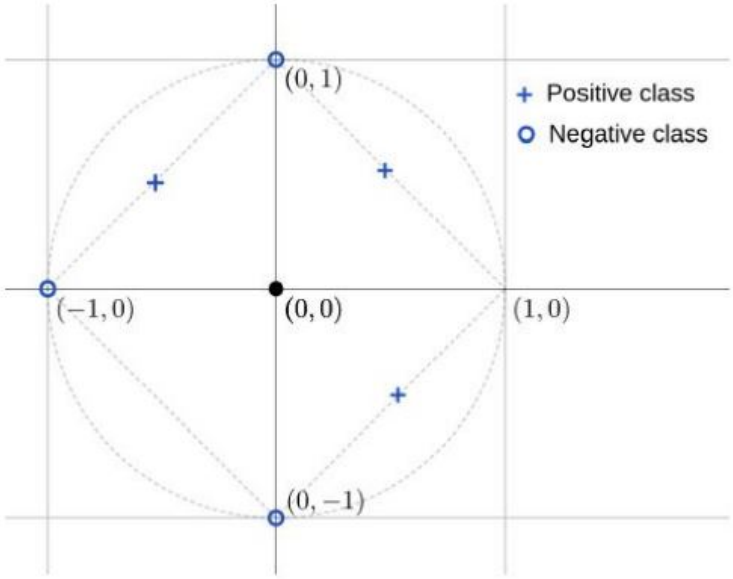
Options :

(a) 4
(b) 5
(c) 9
(d) 12

2. Consider a KNN classifier for a binary classification problem with k = 3. A test-point at (0, 0) has to be classified using this model. The training-dataset is as follows:

- The three negative class data-points are fixed and occupy three vertices of the diamond.
- The three positive class data-points can be anywhere on the edges of the diamond except the vertices

Based on the above data, answer the given sub-questions.



(i) If the Manhattan distance metric is used, what is the predicted label of the test-point?

(a) Positive class

(b) Negative class

(c) It could be either of the two classes. An exact decision requires information about how to break ties.

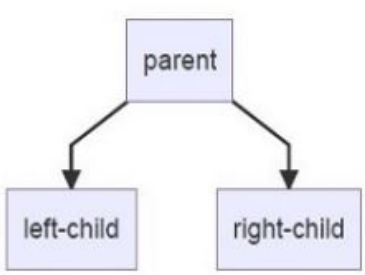
(ii) If the Euclidean distance metric is used, what is the predicted label of the test-point?

(a) Positive class

(b) Negative class

(c) It could be either of the two classes. An exact decision requires information about how to break ties.

3. A decision stump is a decision tree that has exactly one question at the parent node (root) which then splits into two prediction nodes (leaves):



Consider a decision stump for a binary classification problem that has 500 data points at the parent node, out of which 200 data points go into the left child. The number of data points that belong to class 1 in the parent node is 300. The number of data points that belong to class 1 in the left child is 50. The labels are in $\{1, 0\}$.

Note for calculations: Use log_2 for all calculations that involve logarithms. For all questions, enter your answer correct to three decimal places. Use three decimal places even while calculating intermediate quantities.

(i) What is the label assigned to the left child? Enter 1 or 0 .

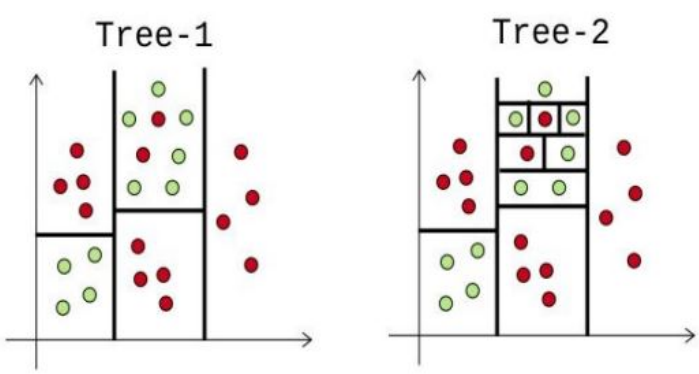
(ii) What is the entropy of the parent?

(iii) What is the entropy of the left child?

(iv) What is the entropy of the right child?

(v) What is the information gain corresponding to the question at the parent node?

4. Consider a training dataset for a binary classification problem in R^2. Two decision trees are trained on the same dataset. The decision regions obtained are plotted for both the trees:



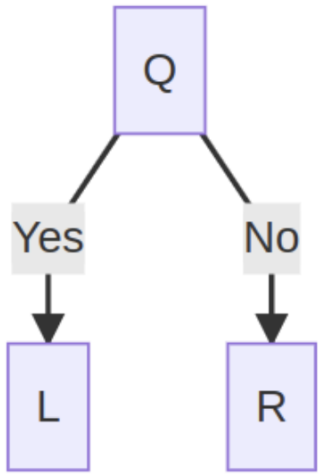
Which of these two trees is likely to perform better on test data.

(a) Tree 1

(b) Tree 2

5. Suppose we have a binary classification dataset with 1000 data points, consisting of 600 points belonging to class 0 and 400 points belonging to class 1. If we use a k-nearest neighbor model with k=900 to predict the class labels of the data points,how many data points will be classified correctly?

6. The dataset at node Q has 1000 points. After asking the question Q it splits into two parts, 250 in node L and 750 in node R.



What is the value of γ? γ is the cardinality of L divided by cardinality of Q?

7. Consider the following statements:

   S1: The impurity of a node is influenced by its ancestors.

   S2: The impurity of a node is influenced by its descendants.

   (a) Only S1 is true
   (b) Only S2 is true
   (c) Both S1 and S2 are true
   (d) Neither S1 nor S2 are true

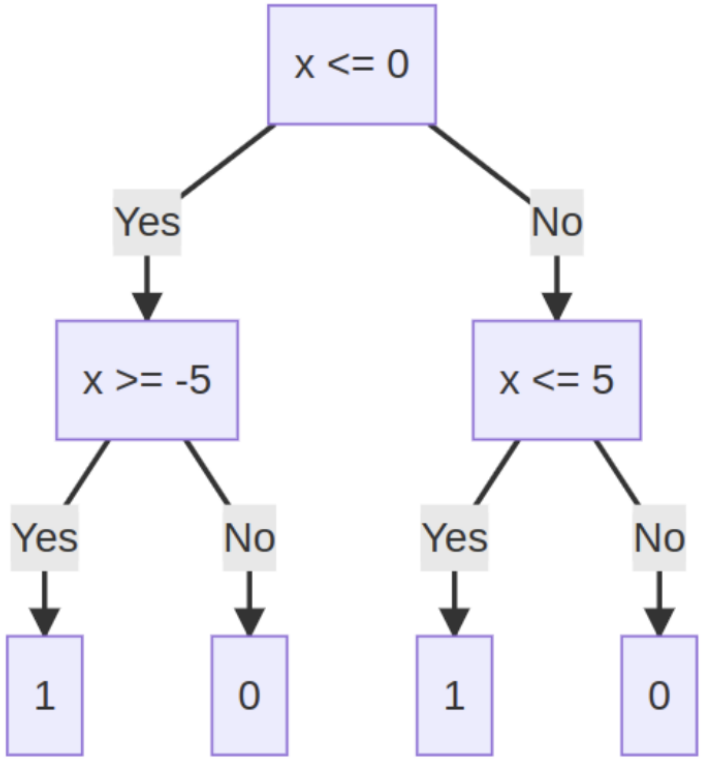
8. Which of the following statements about Decision Trees is true?

   (a) They are only applicable to classification tasks.
   (b) They can handle both categorical and numerical features.
   (c) They are less prone to overfitting compared to other models.
   (d) They are robust to outliers in the training data.

9. Consider the following decision tree for a binary classification problem. The dataset lies in R.

   S_1 is the set of all points in R classified as 1 and S_0 is the set of all points in R classified as 0. How will you describe S_1 and S_0?

   (a) S_1 = [-5,5] and S_0 = [0,5]



(b) S_1 = [-∞,5) and S_0 = (5,∞)

(c) S_1 = [-5,5] and S_0 = (-∞,-5) ⋃ (5,∞)

(d) S_1 = (-∞,-5) ⋃ (5,∞) and S_0 = [-5,5]

10. Which of the following are true about decision trees?

(a) They can be used only for classification.

(b) The tree depth never exceeds O(log n) for n sample points.

(c) All the leaves must be pure.

(d) None of above

Solution:
Q(1) c Q(2) (i)c, (ii)a Q(3) (i)-0, (ii)[0.92-1], (iii)[0.80-0.83], (iv)[0.62-0.68], (v)[0.21-0.29]
Q(4) a, Q(5) 600, Q(6) [0.24,0.26], Q(7) a, Q(8) b, Q(9) c, Q(10) d